

Particulate Matter Monitoring & Health Effects

Issue 2026-08-22 | Entry window 22 Aug 2026 | 13 in scope, 44 screened out

13RECORDS IN SCOPE
(44 SCREENED OUT)**7**OF 10 CLUSTERS
OCCUPIED**6**ABSTRACT-FREE
DEPOSITS (46%)**5**RISK ESTIMATES
ON THE PLOT**3**INFECTION-RELATED
RECORDS

Signal of the day

Three papers, three scales, one claim: particulate exposure degrades the innate pulmonary defence against infection — and today one of them carries a rescue.

Zhang et al. (*Environ Sci Technol*) followed 3,583 hospitalised patients with severe fever with thrombocytopenia syndrome and found long-term pre-admission $PM_{2.5}$ associated with **HR 1.16 (95% CI 1.04–1.30) per $10 \mu g/m^3$** for SFTS-attributable death, with viral load partially mediating; the paired mouse and cell work shows $PM_{2.5}$ pre-exposure raising SFTSV replication and tissue injury, implicates mitochondrial redox and ROS pathways, and *reverses the effect with N-acetylcysteine*. Masci et al. (*Environ Toxicol Pharmacol*) reach the same place from nutrition: in rats, chronic 20% caloric restriction alone reproduced the alveolar-macrophage phenotype that residual oil fly ash produced in well-nourished animals — reduced viability, lower Nrf2, higher $TNF\alpha$ — and undernourished macrophages then **failed to mount a response to a Poly (I:C) viral challenge regardless of PM exposure**, i.e. the two insults are not additive because the first one has already saturated the damage. Introna et al. (*ACS EST Air*) supply the dosimetric caveat that both need: exposing lung cells to subway-platform $PM_{2.5}$ at a realistic *in situ* air-liquid-interface dose of $0.12 \pm 0.07 \mu g/cm^2$ produced **no measurable change in viability or inflammatory markers at 24 h**, while the same particles submerged at laboratory doses moved four cytokines. The uncomfortable synthesis is that the epidemiology is real, the mechanism is plausible, and the standard in-vitro dose is nowhere near the ambient one.

What changed today

- **Six of thirteen records arrived without an abstract — the highest abstract-free share on record.** All six came through the Crossref by-ISSN sweep (*Atmos Environ*, *Atmos Res*, *Atmos Pollut Res*, *Environ Pollut*), and Saturday's thin PubMed leg (2 records) meant the sweep carried 8 of 13. They are shipped as *Metadata only* with design, sample and endpoint left unextracted rather than inferred. Two of them — aerosol pH and preterm birth, and ammonia-driven ROS generation in secondary organic aerosol — are the day's most interesting titles and the least characterisable records in it.
- **A household-fuel transition with a measurable physiological dividend.** In CHARLS, moving from solid to clean cooking fuel attenuated peak-expiratory-flow decline by **11.0 L/min (95% CI 4.7–17.3)** over two years and was associated with **OR 0.80 for chronic lung disease and 0.80 for heart disease** over the long term. This is one of the few PM-relevant exposures where the intervention, the physiological intermediate and the disease endpoint are all measured in the same cohort.
- **An outdoor-activity framing of wildfire smoke, at a two-orders-of-magnitude ratio.** A preprint on the July 2026 Gironde–Landes fires reports the footprint over the EAQI “Poor” threshold (hourly $PM_{2.5} > 50 \mu g/m^3$) at **149 times the burned area**, extending roughly 370 km downwind. The methodological point is the correction the author makes against himself: the 10 km CAMS analysis reproduces the episode (daily $r = 0.83$) but underestimates peaks by a median factor of ~ 2.2 , so the mapped heat–smoke relay is a conservative floor, not an estimate.
- **Childhood eczema, with traffic proximity separated from the household cluster.** A 1,227-child Porto survey isolates **aOR 1.32 (1.11–1.48) for residence within 300 m of a traffic zone** alongside **aOR 1.48 (1.26–1.73) for postnatal secondhand smoke**, after adjustment that also absorbs family history (aOR 3.53), income and maternal education. Cross-sectional and questionnaire-based — but the traffic term surviving that adjustment set is the part worth citing.
- **Front-matter contamination in the newly promoted ES&T feed, caught and closed.** The first run with the widened ISSN list pulled **154 works from *Environ Sci Technol***, of which **142 were mastheads and issue-information stubs** from volumes 42–52 being re-deposited in bulk. They carry DOIs and titles, so nothing downstream would have rejected them. A title-pattern filter now drops them at the source.

Corpus analytics

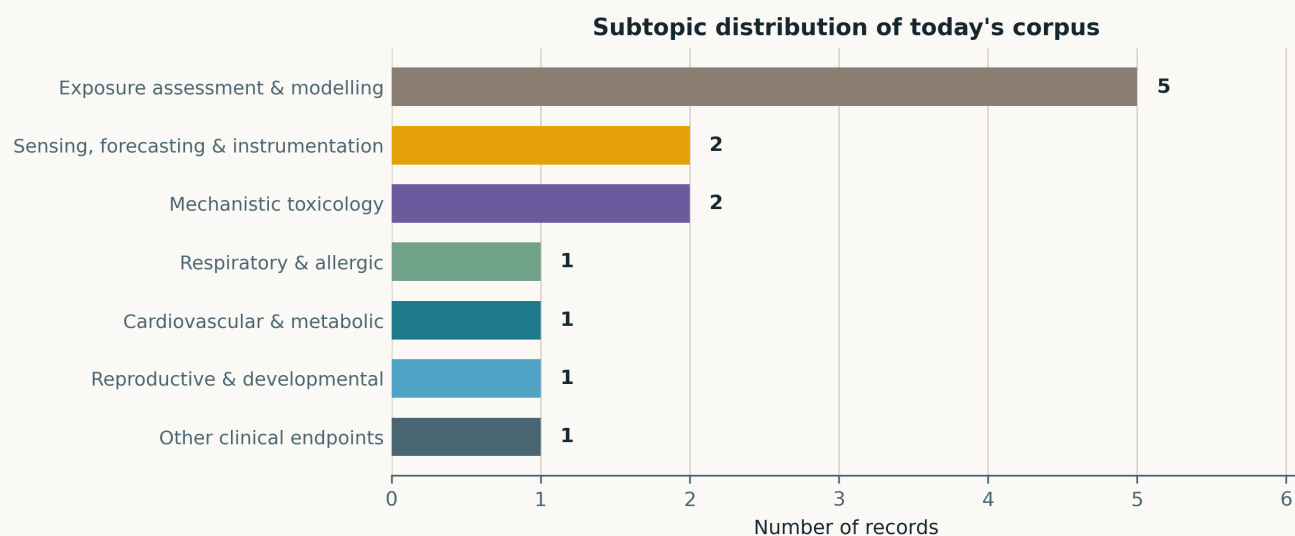


Fig. Subtopic assignment of the 13 in-scope records; single-label by dominant contribution. *Exposure assessment & modelling* takes 5, and the health side is unusually evenly spread — mechanistic toxicology 2, and one record each in cardiovascular/metabolic, respiratory/allergic, reproductive/developmental and other clinical endpoints. Seven of ten canonical clusters are occupied on a 13-record day, which is a wider spread than the 19-record issue that preceded it.

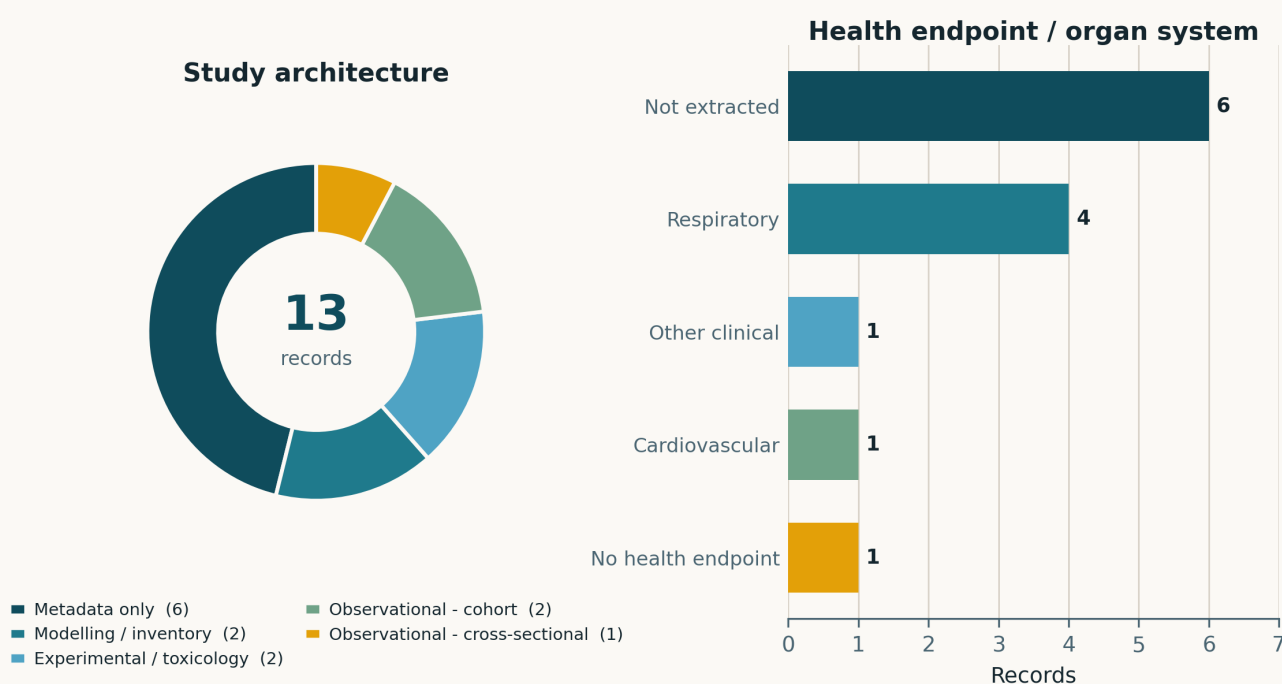


Fig. Left — study architecture. *Metadata only* is the largest single slice at 6 of 13 (46%), the highest share the watch has recorded. That slice is a statement about what publishers deposited, not about what was published: every one of the six is a Crossref journal-sweep record whose publisher supplied a title and DOI but no abstract. Right — health endpoint by organ system. Four records carry a respiratory endpoint, and the *Not extracted* bar is exactly the six abstract-free records.

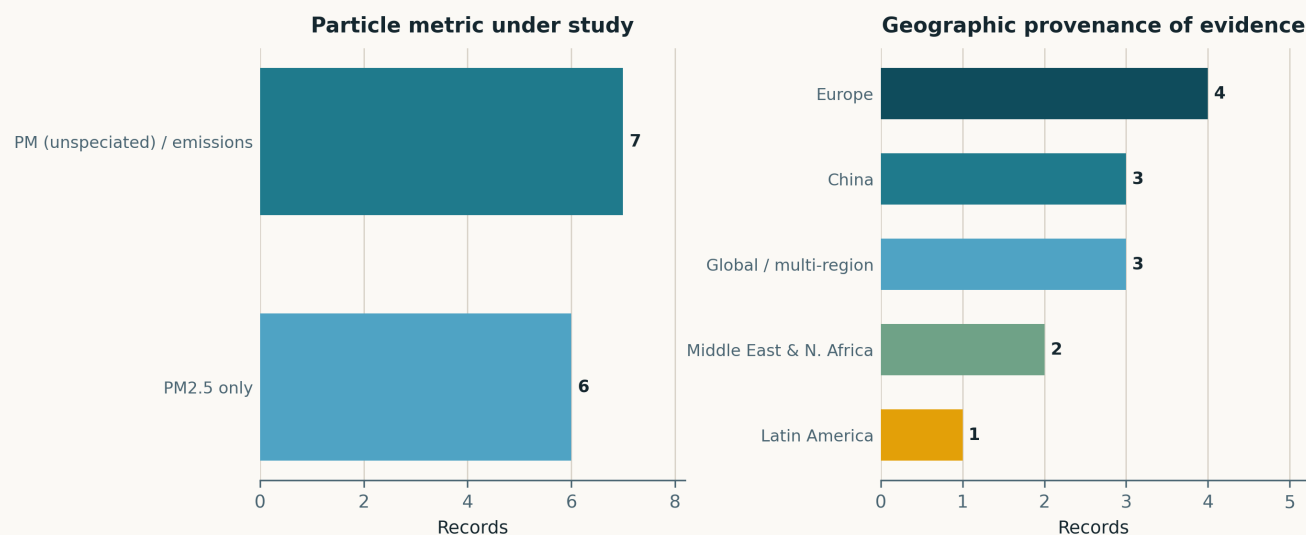


Fig. Left — particle metric. Six records are PM_{2.5}-specific and seven are unspecified PM or a PM-proxy exposure (household solid fuel, traffic proximity, ROFA, mineral dust, aerosol acidity). Right — geographic provenance. Europe leads with 4, China with 3, and the Middle East / North Africa contributes 2 — an Iranian winter-PM_{2.5} synoptic study and a Mesopotamian dust projection, both metadata-only. The three “global / multi-region” records include two whose study location is genuinely not stated in the deposited metadata.

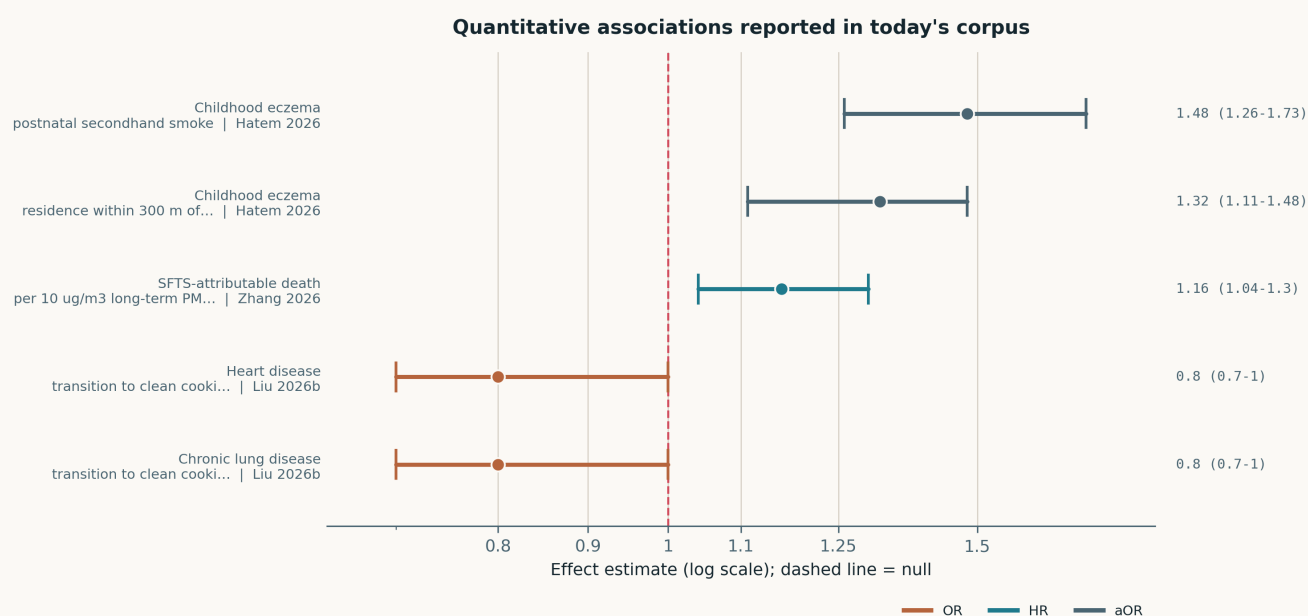


Fig. All five forest-plottable estimates, on a common logarithmic axis. **The exposure contrasts are not mutually comparable** — a per-10 $\mu\text{g}/\text{m}^3$ ambient PM_{2.5} contrast, two binary household/residential exposures, and a fuel-transition contrast — so read direction and precision, not relative magnitude. The two Liu et al. odds ratios are protective and their published intervals are given to one decimal (0.7–1.0), so the upper bound touches unity even though the paper reports $p = 0.044$ and $p = 0.028$; they are transcribed as published rather than back-computed from the p -values.

Life-course windows addressed today (records may span several stages)



Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored at that stage. Four of five windows are occupied — the widest life-course coverage since 14 August — and the day spans the whole arc from an aerosol-acidity/preterm-birth record through a paediatric eczema survey to a middle-aged-and-older cohort. Only adolescence is empty.

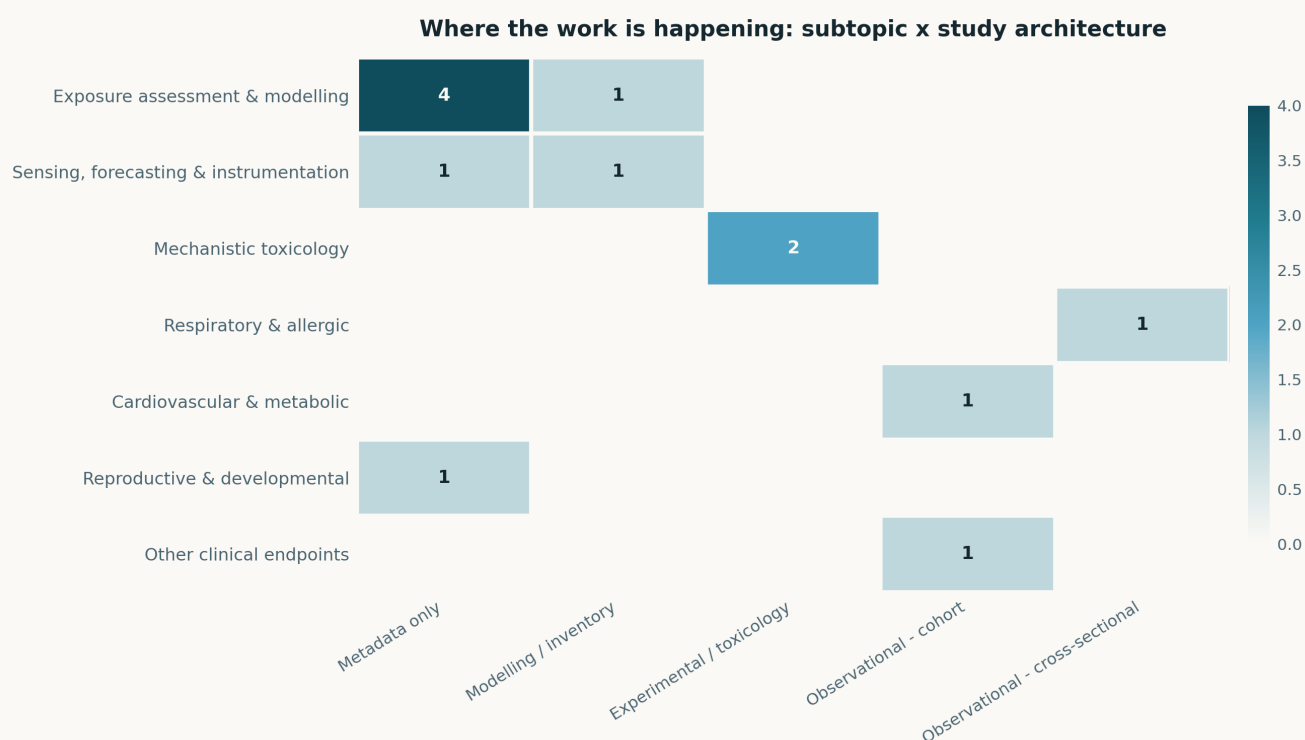


Fig. Subtopic × study-architecture cross-tabulation. The concentration of the *Metadata only* column across four different subtopic rows is the structural feature of the day: the abstract-free deposits are not clustered in one corner of the corpus, so the gap they leave is spread across the whole issue rather than confined to one topic.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Mechanistic toxicology

2 records

Introna et al. 2026 ACS EST Air

Novel approach to assess the toxicity of PM_{2.5} on a subway platform using a mobile air-liquid interface system and submerged exposure

A mobile ALI exposure system operated *on the platform itself*, with A549 monoculture and A549 + dTHP-1 coculture, benchmarked against conventional submerged laboratory exposure to the same metal-rich mechanical-wear particles. At the realistic in-situ ALI dose of $0.12 \pm 0.07 \mu\text{g}/\text{cm}^2$ there was **no change in viability or inflammatory markers at 24 h**. Submerged at higher dose, viability fell in both cell types; A549 released IL-6 and TNF α at 100 $\mu\text{g}/\text{mL}$, and dTHP-1 released IL-8 at the lowest dose tested (10 $\mu\text{g}/\text{mL}$) with all four cytokines elevated at 100. The authors' own reading is the right one — longer in-situ exposures are needed to reach comparable dose — and the wider implication is that the submerged literature's dose range is not the ambient one.

doi: 10.1021/acsestair.6c00018

Masci et al. 2026 Environ Toxicol Pharmacol

Air pollution and undernutrition induce rat alveolar macrophage functional dysregulation and compromise their response to a viral challenge

Male Wistar rats, control vs. nutritional growth retardation (20% caloric restriction, 4 weeks), each with intranasal residual oil fly ash or vehicle; isolated alveolar macrophages assayed for viability, TNF α , oxidative stress, Nrf2 and LC3, then challenged with Poly(I:C). Undernutrition alone reproduced what ROFA did to well-nourished animals. **ROFA added little or nothing on top of undernutrition** — an attenuated rather than additive response — and after Poly(I:C), control macrophages mounted an inflammatory response that prior ROFA *amplified*, while undernourished macrophages failed to respond at all regardless of exposure. The interaction structure, not the main effect, is the finding: in a population that is already nutritionally compromised, the marginal immunological cost of PM may be small because the defence is already gone.

doi: 10.1016/j.etap.2026.105138

Other clinical endpoints

1 record

Zhang et al. 2026 Environ Sci Technol*PM_{2.5} exposure and the prognosis of severe fever with thrombocytopenia syndrome*

A cohort of **3,583 hospitalised SFTS patients** with long-term pre-admission ambient PM_{2.5}: **HR 1.16 (95% CI 1.04–1.30) per 10 µg/m³** for SFTS-attributable death, with exploratory mediation implicating viral load. A counterfactual burden assessment estimates **137 avoidable SFTS deaths in Xinyang, 2010–2022** had annual PM_{2.5} met the Chinese national standard. The experimental arm shows PM_{2.5} pre-exposure increasing SFTSV replication and tissue injury; transcriptomics implicate mitochondrial redox and ROS pathways, and ROS/MDA measurement plus N-acetylcysteine intervention support oxidative priming as the upstream step. Emerging-infection prognosis is a genuinely new endpoint class for this watch.

doi: 10.1021/acs.est.6c03214

Cardiovascular & metabolic**1 record****Liu et al. 2026b** Indoor Air*Household clean cooking fuel transition and attenuated cardiopulmonary function decline in middle-aged and older adults*

CHARLS 2011–2020, with two independent subcohorts built from baseline solid-fuel users: a short-term physiological cohort (n = 6,534; 27.4% transitioned by 2013) and a long-term disease cohort (n = 6,362; 33.5% transitioned by 2015). Transition attenuated peak-expiratory-flow decline by **11.0 L/min (95% CI 4.7–17.3)** and lowered systolic blood pressure modestly ($\beta = -1.7$, $p = 0.017$). Long term: **OR 0.80 (0.7–1.0, $p = 0.044$) for chronic lung disease** and **0.80 (0.7–1.0, $p = 0.028$) for heart disease**, with no association for hypertension; protective effects were larger in non-smokers and non-drinkers. The two-subcohort design is what lets the physiological intermediate and the disease endpoint be read together.

doi: 10.1155/ina/8898458

Respiratory & allergic**1 record****Hatem et al. 2026** Eur J Pediatr*Child characteristics, household and home environmental factors associated with eczema*

1,227 children aged 5–11 from 20 primary schools in Porto (46.5% response), analysed with mixed-effects logistic regression with a school random intercept and multiple imputation. Independently associated with higher odds: **residence within 300 m of a traffic zone, aOR 1.32 (1.11–1.48); postnatal secondhand smoke, aOR 1.48 (1.26–1.73)**; family history 3.53; higher family income 1.40; all-purpose cleaner use 1.16. Lower odds: maternal education 0.68, detached housing 0.67, fur blanket use 0.85. Household dampness interacted with residence type and winter ventilation frequency. Cross-sectional and self-reported, so temporality is unavailable — but the traffic term surviving a socioeconomic adjustment set this dense is the citable part.

doi: 10.1007/s00431-026-07327-1

Reproductive & developmental**1 record****Han et al. 2026** Environ Pollut*Aerosol pH associated with preterm birth risk: unmasking the hidden role of atmospheric acidity in maternal-fetal health*

Metadata only. Surfaced through the Crossref by-ISSN sweep with no abstract deposited; title, journal, authorship and DOI verified, nothing else extracted. Worth flagging by title alone: aerosol acidity as an exposure metric in a reproductive endpoint is not something this watch has seen before, and it pairs directly with Chai et al.'s pH-governed partitioning result in yesterday's issue. Queued for abstract retry.

doi: 10.1016/j.j.envpol.2026.129011

Sensing, forecasting & instrumentation

2 records

Fei et al. 2026 Front Immunol

SHAP-interpretable machine learning integrating exposures and multi-omics in COPD-lung cancer comorbidity

A sequential cross-scale framework: GBD epidemiology 1990–2021 across 204 countries, exposome-wide random-forest classification with SHAP interpretation, immune characterisation via CIBERSORT and single-cell RNA sequencing, candidate-gene prioritisation by SMR and LASSO, then wet-lab validation including a macrophage-epithelial Transwell co-culture. **SHAP identified smoking, particulate matter pollution and residential radon as the major shared risk factors**; TREM1 and ODF2L emerged as shared hub genes, downregulated in both diseases, and macrophage-specific silencing of either attenuated macrophage-driven A549 migration and proliferation. The exposure side is GBD-level and coarse — this is not an exposure-assessment paper — but it is a clean worked example of putting an exposome variable through an interpretable-ML pipeline and ending at a testable target.

doi: 10.3389/fimmu.2026.1875146

Zeng et al. 2026 Atmos Environ

Aerosol vertical distributions determine the seasonal relationships between column AOD and surface PM_{2.5} over Europe

Metadata only. No abstract deposited. The title states a result that matters directly to satellite-based PM_{2.5} estimation — the AOD-to-surface conversion being governed by vertical structure and therefore seasonally non-stationary — so this is the highest-priority record of the six for abstract retry or full-text retrieval on the next run.

doi: 10.1016/j.atmosenv.2026.122305

Exposure assessment & modelling

5 records

Defrance 2026 (preprint) Research Square

Beyond the fire perimeter: the smoke footprint that constrains outdoor activity dwarfs the burned area of the July 2026 Gironde–Landes wildfire episode

ERA5-Land wet-bulb globe temperature combined with CAMS analysis PM_{2.5} at hourly resolution over 25–31 July 2026, against 40 background monitoring stations. The two fires burned 340 km² and forced ~220,000 evacuations; the footprint above the European AQI “Poor” threshold (hourly PM_{2.5} > 50 µg/m³, where reduced outdoor activity is advised) reached **149 times the burned area** and about **370 km downwind**, with 139× on the EU 2030 daily limit of 25 µg/m³ and at least 121× against the larger operational fire perimeter. Plume co-location rose from 73% hourly to 94% daily. At the stations, a same-day heat–smoke relay affected 28% of affected station-days at 25 µg/m³ and 19% at 50, against 0–2% in the co-located grid cells — **the 10 km analysis smooths away the peaks that drive the relay**, and reproduces the episode at daily $r = 0.83$ while underestimating peaks by a median factor of ~2.2. Preprint, single author, single episode; the self-directed error analysis is unusually thorough for the format.

doi: 10.21203/rs.3.rs-10757961/v1

Qing et al. 2026 Atmos Pollut Res

Seasonal and urban-suburban characteristics, emission and spatial sources of PM_{2.5}-bound PAHs and n-alkanes in Wuhu City

Metadata only. Crossref sweep, no abstract deposited. A source-apportionment study of the particle-bound organic fraction, which is the composition class the previous issue’s mortality result pointed at — flagged for retry rather than characterised here.

doi: 10.1016/j.apr.2026.103186

Liu et al. 2026 Atmos Res

Identification of dominant weather patterns and circulation regulation mechanisms of winter PM_{2.5} pollution in Iran

Metadata only. Crossref sweep, no abstract deposited. Synoptic classification of winter PM_{2.5} episodes; one of very few Iranian PM records the watch has surfaced, which is why it ships as a stub rather than being dropped.

doi: 10.1016/j.atmosres.2026.109287

Abadi et al. 2026 Atmos Res

Future projections of surface dust concentrations in major dust source regions of the Mesopotamian plains

Metadata only. Crossref sweep, no abstract deposited. Projected mineral-dust concentrations for a source region that exports coarse PM across the Gulf and into South Asia; no scenario, horizon or model family is extractable from the deposited metadata.

doi: 10.1016/j.atmosres.2026.109282

Cai et al. 2026 Atmos Environ

Effect of ammonia on reactive oxygen species generation in secondary organic aerosols

Metadata only. Crossref sweep, no abstract deposited, and no study location stated. Ammonia-modulated oxidative potential of secondary organic aerosol sits precisely between the previous issue’s Nanjing ammonia-thermodynamics result and today’s oxidative-priming mechanism papers, so it is the second-highest-priority retry after Zeng et al.

doi: 10.1016/j.atmosenv.2026.122307

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Liu et al.	Household clean cooking fuel transition and attenuated cardiopulmonary function decline: a national longitudinal study of middle-aged and older adults in China	Indoor Air	Cohort	Household solid-fuel to clean-fuel cooking transition	China
Reproductive & developmental					

First author	Focus	Journal	Design	Metric	Region
Han et al.	Aerosol pH associated with preterm birth risk: unmasking the hidden role of atmospheric acidity in maternal-fetal health	<i>Environ Pollut</i>	Metadata only (no abstract)	Aerosol acidity (pH) (not extractable)	Not stated
Respiratory & allergic					
Hatem et al.	Child characteristics, household, and home environmental factors associated with eczema: a cross-sectional study	<i>Eur J Pediatr</i>	Cross-sectional survey	Traffic-zone proximity (<300 m) and postnatal secondhand smoke	Porto, Portugal
Mechanistic toxicology					
Introna et al.	Novel approach to assess the toxicity of PM _{2.5} on a subway platform using a mobile air-liquid interface system and submerged exposure	<i>ACS EST Air</i>	In vitro exposure	Subway-platform PM _{2.5} , metal-rich; ALI dose 0.12 +/- 0.07 ug/cm2	Sweden (Stockholm Univ. affiliation; platform city not stated)
Masci et al.	Air pollution and undernutrition induce rat alveolar macrophage functional dysregulation and compromise their response to a viral challenge	<i>Environ Toxicol Pharmacol</i>	Animal experiment	Residual oil fly ash (ROFA), intranasal, sub-chronic	Argentina
Sensing, forecasting & instrumentation					
Fei et al.	SHAP-interpretable machine learning integrating exposures and multi-omics reveals immune alterations and biomarkers in COPD-lung cancer comorbidity	<i>Front Immunol</i>	Machine-learning model	Particulate-matter pollution as a GBD exposure risk factor	Global
Zeng et al.	Aerosol vertical distributions determine the seasonal relationships between column AOD and surface PM _{2.5} over Europe	<i>Atmos Environ</i>	Metadata only (no abstract)	Column AOD vs surface PM _{2.5} (not extractable)	Europe
Exposure assessment & modelling					
Abadi et al.	Future projections of surface dust concentrations in major dust source regions of the Mesopotamian plains	<i>Atmos Res</i>	Metadata only (no abstract)	Projected surface mineral dust (not extractable)	Iraq
Cai et al.	Effect of ammonia on reactive oxygen species generation in secondary organic aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	SOA reactive oxygen species yield (not extractable)	Not stated
Defrance 2026	Beyond the fire perimeter: the smoke footprint that constrains outdoor activity dwarfs the burned area of the July 2026 Gironde-Landes wildfire episode	<i>Research Square</i>	Modelling / inventory	CAMS analysis PM _{2.5} + 40 background monitors	Gironde-Landes, France
Liu et al.	Identification of dominant weather patterns and circulation regulation mechanisms of winter PM _{2.5} pollution in Iran	<i>Atmos Res</i>	Metadata only (no abstract)	Winter PM _{2.5} under synoptic weather types (not extractable)	Iran
Qing et al.	Seasonal and urban-suburban characteristics, emission and spatial sources of PM _{2.5} -bound PAHs and n-alkanes in Wuhu City	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} -bound PAHs and n-alkanes (not extractable)	Wuhu, China
Other clinical endpoints					
Zhang et al.	PM _{2.5} exposure and the prognosis of severe fever with thrombocytopenia syndrome: an integrated epidemiological and experimental study	<i>Environ Sci Technol</i>	Cohort	Long-term pre-admission ambient PM _{2.5}	Xinyang, China

Provenance and method

Entry window **2026-08-22** → **2026-08-22**, contiguous with the 21 August issue. Built at 22:4x local on 22 August, satisfying the 22:00 rule for same-day harvesting. **Sources.** PubMed E-utilities on [EDAT] returned only **2 records**, both shipped — a Saturday floor. Europe PMC on CREATION_DATE returned 23 unique records, of which **15 were archival PMC back-scans** (legacy scans, no title, publication year decades outside the window), 3 were already seen or belonged to the 21 August issue, and 3 shipped. The Crossref by-ISSN sweep returned 36 records after front-matter filtering and **shipped 8 — 62% of the issue. OpenAlex returned HTTP 429 for the 20th consecutive run.** arXiv returned nothing new under the 3-day screen. **Two harvester changes this run.** The supplementary ISSN list trialled on 20 August was promoted into `harvest.JOURNALS` (15 additions, each ISSN verified against Crossref's journal index; MDPI *Sensors* deliberately excluded after returning 27 deposits for 0 in-scope records). The first run with *Environ Sci Technol* in the list returned 154 works of which **142 were mastheads and issue-information stubs** being re-deposited in bulk from volumes 42–52; a title-pattern front-matter filter now drops these at the source, and the count is logged per journal. **Screening.** 44 records were screened out and written to `state/rejected.jsonl` with an individual reason each. Two borderline exclusions are logged as `borderline`: combustion SVOC residue on wildland-firefighter protective gear, and droplet-residue resuspension from medical PPE — both occupational and combustion-adjacent, neither carrying an airborne particulate measurement. **Labels.** Six records carry no abstract and are marked *Metadata only*; their design, sample size, endpoint and (in two cases) study location are recorded as not extracted rather than inferred. Effect estimates are transcribed verbatim from abstracts. All DOIs were checked by `check_dois.py` before publication. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.